

PERIPHERAL VASCULAR DISEASE (PVD)

- Peripheral vascular disease is characterised by claudication: muscle pain in legs or buttocks on exercise that is relieved by rest. Leg pulses are reduced and skin may be cool, shiny and hairless.
- Refer the patient newly diagnosed with peripheral vascular disease for specialist assessment.

Give urgent attention to the patient with peripheral vascular disease and any of:

- Sudden severe leg pain at rest with any of the following in the leg: numbness, weakness, pallor, no pulse: **acute limb ischaemia** likely
- Leg pain occurring at rest, ulcer or gangrene on leg: **critical limb ischaemia** likely
- Pulsatile mass in abdomen with abdominal/back pain or BP < 90/60: **ruptured abdominal aortic aneurysm** likely

Management:

- Acute limb ischaemia** likely: refer urgently.
- Critical limb ischaemia** likely: discuss same day urgency of referral with specialist.
- Ruptured abdominal aortic aneurysm** likely: avoid giving IV fluids even if BP < 90/60 (raising blood pressure may worsen the rupture) and refer urgently.

Assess the patient with peripheral vascular disease

Assess	When to assess	Note
Symptoms	Every visit	- Manage symptoms as on symptom pages. Ask about chest pain ↗ 137 and symptoms of stroke/TIA ↗ 136. - Document the walking distance before onset of claudication.
Depression	Every visit	In the past month, has patient: 1) felt down, depressed, hopeless or 2) felt little interest or pleasure in doing things? If yes to either ↗ 143.
BP	Every visit	If known hypertension ↗ 133. If not, check BP: if $\geq 140/90$ ↗ 132.
Legs and feet	Every visit	Check for pain, pulses, sensation, deformity, skin problems. For foot screen and foot care education ↗ 66.
Abdomen	Every visit	If a pulsatile mass felt, refer for assessment for possible abdominal aortic aneurysm. Refer urgently if abdominal/back pain or BP < 90/60.
Diabetes risk	At diagnosis, then yearly	If known diabetes ↗ 130. If not known with diabetes, check glucose ↗ 17.

Advise the patient with peripheral vascular disease

- Advise the patient to keep legs warm and below heart level (especially at night), and to avoid decongestant medications that may constrict blood vessels.
- If patient smokes, encourage to stop ↗ 141. Support the patient to make a change ↗ 177.
- Advise patient that physical activity is an important part of treatment. It increases the blood supply to the legs and may significantly improve symptoms.
- If < 55 years (man) or < 65 years (woman), advise the first degree relatives (parents, siblings, children) to have CVD risk assessment ↗ 127.

Treat the patient with peripheral vascular disease

- Help the patient to manage his/her CVD risk ↗ 129. Advise brisk exercise for 30 minutes at least 3 times a week (preferably daily). Advise patient to pause and rest whenever claudication develops.
- Give **simvastatin**¹ 40mg² at night regardless of cholesterol level. If patient develops muscle pain/cramps, reduce dose to 10mg daily. Avoid if pregnant or liver disease.
- Give **aspirin** 150mg daily for life. Avoid if peptic ulcer, dyspepsia, kidney or liver disease.

- Refer to specialist at diagnosis (start medications and exercise while waiting for appointment) and if pain interferes with activities of daily living after 3 months of medication and exercise.
- Review 3 monthly until stable (coping with activities of daily living and able to work), then yearly.**

¹If on lopinavir/ritonavir or atazanavir/ritonavir, avoid simvastatin, give instead **atorvastatin** 10mg at night. ²If on amlodipine, reduce simvastatin dose to 20mg at night. No dose adjustment needed for rosuvastatin, pravastatin, atorvastatin.